

# CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor of Computer Applications(BCA)  
Semester-I University Examination November-2019

Date: 12.11.19 , Tuesday  
Time:10.00a.m.to 01.00 p.m.

## INTRODUCTION TO INFORMATION TECHNOLOGY [CA109]

**Marks: 70**

**Duration: 180 mins.**

### SECTION-I

**Answer all the questions.**

Section Duration: 30 mins

Make suitable assumptions when necessary.  
Simple Calculator is permitted.

- Q.1) Salary of an employee is considered as which kind of data? (1)  
1) [Text](#)[Numeric](#)[Image](#)[None of Above](#)
- 2) Input system of computer is used to \_\_\_\_\_data from the external world (1)  
environment.  
[Store](#)[Interpret](#)[Execute](#)[Acquire](#)
- 3) A diamond indicates which action in the flow chart? (1)  
[Decision](#)[Process](#)[Output](#)[None of the above](#)
- 4) An \_\_\_\_\_is the signal to the processor emitted by hardware or software that (1)  
needs immediate attention.  
[Interface](#)[Inquiry](#)[Interaction](#)[Interrupt](#)
- 5) What is the hexadecimal value of 5678 decimal number? (1)  
[162F](#)[162E](#)[1621](#)[162D](#)
- 6) \_\_\_\_\_is the octal value of the binary number 100010. (1)  
[344](#)[224](#)[None of the Above](#)
- 7) Video signal in the form of sampler is converted in digital form by \_\_\_\_\_. (1)  
[Encoder](#)[Analog Converter](#)[Faltbed](#)[None of the Above](#)
- 8) \_\_\_\_\_is not the characteristic of the secondary storage. (1)  
[Volatile](#)[Portable](#)[\(1\) and \(2\) both](#)[None of the Above](#)
- 9) \_\_\_\_\_is non volatile in nature. (1)  
[ROM](#)[SRAM](#)[DRAM](#)[\(2\) and \(3\) both](#)
- 10) \_\_\_\_\_storage cell is used for laser disk or compact disk. (1)  
[Flipflop](#)[Capacitor](#)[Magnetic](#)[Polycarbonate](#)
- 11) What is the main purpose of CPU in computer? (1)  
[Acquire Instructions](#)[Store instructions](#)[Interpret and Execute instructions](#)[All Above](#)
- 12) Compare and Bit Operators are the examples of \_\_\_\_\_type of operation. (1)

Input/OutputMemoryLogicalControl

- 13) Which one is not an instruction of Input/Output operation? (1)  
READLOADDISPRT
- 14) \_\_\_ instruction is used to "Take Number from address A and put it in CPU register". (1)  
READSTOLOADDIS
- 15) What is the full form of EDP? (1)  
Electronic Data Processing    Electrical Digital Processing    Electronic Digital Processing    Electrical Data Preprocessing
- 16) Online Chatting is an example of \_\_\_\_\_real time system. (1)  
HardSoftAverageNone of the Above
- 17) Almost every software applications are considered as \_\_\_\_\_applications. (1)  
DesktopOnlineBusinessMobile
- 18) Industry 4.0 is also known as \_\_\_\_\_. (1)  
Smart FactoryDigital 4.0Revolutionary FactoryIoT 4.0
- 19) IR light is similar to visible light except that it has \_\_\_\_\_weavelength. (1)  
ShorterLongerNormalComplex
- 20) \_\_\_\_\_security architecture having flexible security measures. (1)  
IntegratedAdaptiveInfrastructureAcquired

**SECTION-II**

**Answer all the questions.**

Make suitable assumptions when necessary.

Simple Calculator is permitted.

- Q.2) What do you mean by Information Technology? Discuss different types of data with (5)  
A) suitable example. Differentiate Data and Information.
- [OR] What is Flow Chart? Discuss different symbols of it. Draw a Flow Chart to determine (5)  
B) maximum number from any three numbers.
- C) Explain ASCII, ISCII, UTF-8 and UTF-16 standards to represent characters in (4)  
computer.
- [OR] Convert following decimal numbers to binary : (4)  
D) (1) 988 (2) 756
- E) What is the purpose of Random Access Memory in computer? Differentiate DRAM (4)  
and SRAM.
- [OR] Explain OCR, MICR, Bar Code Reader and OMR input devices with proper (4)  
F) example.
- Q.3) Explain different types of ROM. (4)  
A)
- [OR] Define : image scanner. Discuss different parts of flatbed scanner. (4)  
B)
- C) Explain data acquisition steps. (3)

- [OR] Convert 1100111 binary number to decimal number. (3)  
 D)  
 E) Explain memory and processing systems of computer. Discuss steps of data processing by computer. (5)  
 [OR] Give characteristics of storage cell. Discuss advantages and disadvantages of (5)  
 F) Flipflop storage cell.

### SECTION-III

**Answer all the questions.**

Make suitable assumptions when necessary.

Simple Calculator is permitted.

- Q.4) Explain different categories of software applications with proper example. (5)  
 A)  
 [OR] Explain atleast ten functions of spread sheet software. (5)  
 B)  
 C) Explain arithmetic and memory based instruction sets of CPU. Write a CPU interpretation program of adding two numbers. (4)  
 [OR] Give advantages and elements of EDP. (4)  
 D)  
 E) Explain Bluetooth with advantages. (3)  
 [OR] Define following terms : (3)  
 F) (1) Internet of Things (2) Big Data Analytics (3) Virtual Reality  
 Q.5) Explain atleast twelve societal benefits of WWW. (6)  
 A)  
 [OR] Write a problem statement of online banking system based on your thinking (6)  
 B) process. Discuss at least six features of it based on your problem statement.  
 C) Explain GPS. List out any four applications of GPS. (3)  
 D) Explain features of Hostel Management system. (4)  
 [OR] Give the definition of following : (4)  
 E) (1) E-Commerce (2) Smart Card (3) IR Communication (4) Intellectual Property

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